Asking the Right Questions: Benchmarking Large Language Models in the Development of Clinical Consultation Templates

Liam G. McCoy[1,2,3]†, Fateme Nateghi Haredasht[4]†, David Wu[5], David JH Wu[4], Sarita Khemani[4], Vishnu Ravi[4], Saloni Kumar Maharaj[4], Kelvin Zhenghao Li[4,7,8], Abass Conteh[4], Yingjie Weng[4], Kanav Chopra[4], Daniel Shirvani[6], Leah Rosengaus[4], Arth Pahwa[1], Lena Giang[4], Olivia Jee[4], Ethan Goh[4], Jonathan H. Chen*[4]*

[1] University of Alberta; [2] Beth Israel Deaconess Medical Center; [3] Massachusetts Institute of Technology; [4] Stanford (affiliations include Stanford University, Stanford School of Medicine, and Stanford Health Care) [5] Harvard Medical School / Mass General Brigham; [6] University of British Columbia; [7] Tan Tock Seng Hospital; [8] Nanyang Technological University. †equal contribution; * jointly supervised; Corresponding author: Ethan Goh (ethangoh@stanford.edu)

The Challenge:

What are E-Consults at Stanford? An asynchronous, EHR-embedded system for generalists to request specialist input on challenging cases using structured templates.

Why E-Consult Templates? Because follow-up is limited, templates ensure key information is collected consistently.

Why Automated Template Generation? To automate consults, models must be able to gather the right information from consulting physicians. The e-consult setting provides both a practical workflow question, and a unique benchmark of model metacognition and prioritization.

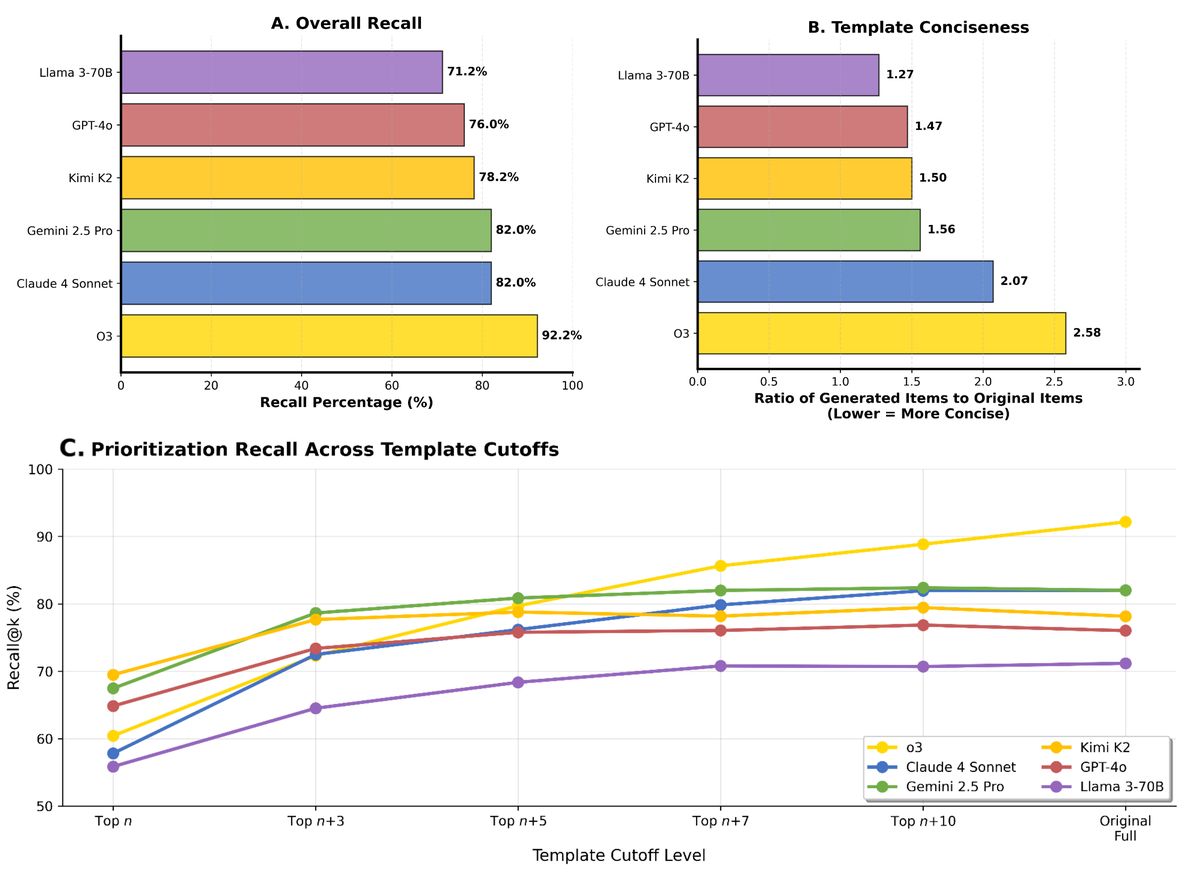

Figure 2. Benchmarking model performance on the template reference set. (A) Recall (% of expert items recovered). (B) Conciseness (generated ÷ original expert items, lower is better). (C) Recall@k under size constraints using model-based prioritization (k = generated template length, n = length of original template).

What We Found:

High comprehensiveness, poor conciseness: Models were thorough but verbose: o3 covered 92% of items but produced ~2.6× more fields (*Fig. 2A/B*). Verbosity and coverage correlated.

Models struggled with effective clinical prioritization: Under size limits, recall fell sharply (o3: 92%→60%; Kimi: 78%→70%), revealing weak prioritization *(Fig. 2C)*.

Marked specialty heterogeneity: Performance varied widely: structured specialties showed >90% recall, whereas narrative fields fell below 70% (often <45% at top-n).

Similar problems persist for novel template generation: For novel topics, clinicians judged 96% of included items appropriate but only 73% necessary, adding ~1 item per template. Specialty-specific heterogeneity persisted *(Fig. 3)*.

Our Conclusions:

- **LLMs can generate clinically comprehensive templates, but conciseness and prioritization remain limitations.**
- **Further work is necessary to ensure such systems reduce (rather than redistribute) the cognitive burden of specialist consultation for both physicians involved.**

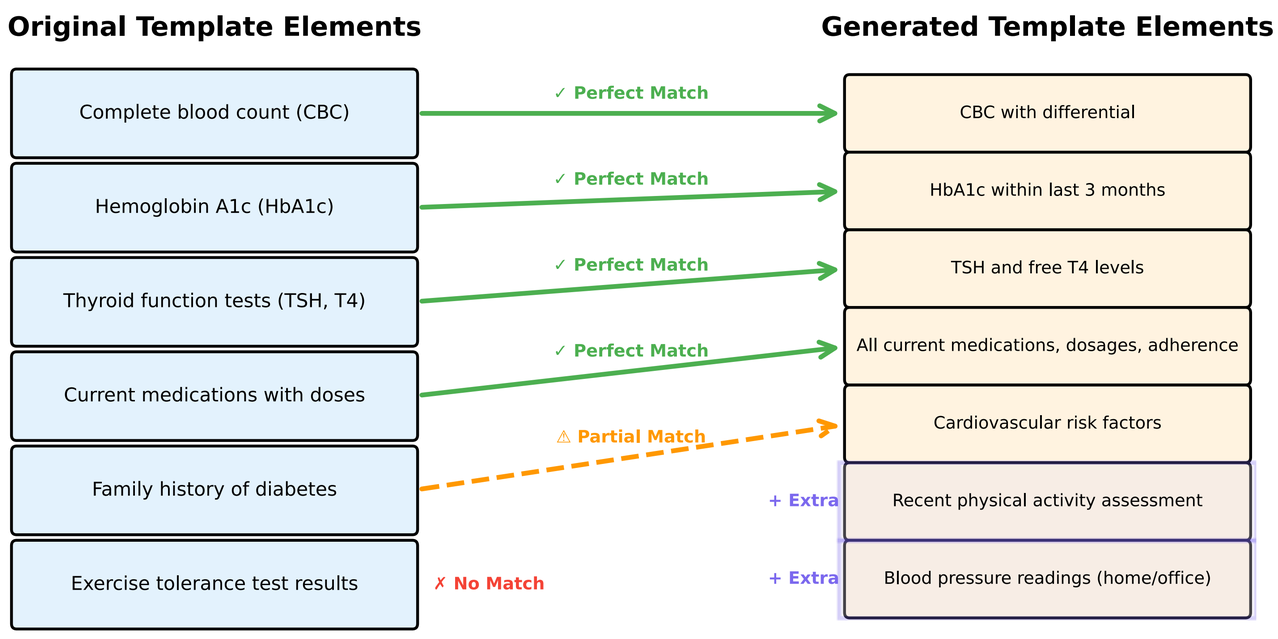

Figure 1. Schematic representation of semantic template matching process. Example alignment of expert-template elements to a model-generated template for diabetes mellitus. Arrows represent autograder matches, with color representing human rater assessment of full vs partial match.

What We Did:

Autograder Validation: Validated an o3-based autograder with judgment of physicians (94% full, 5% partial agreement) (*Fig. 1*)

Template Generation: Used DSPy auto-prompting on 20 training templates; evaluated generation on 125 reference templates.

Model Benchmarking: Benchmarked six LLMs for recall, conciseness, and prioritization via item-ranking at multiple cutoffs.

Novel Template Generation: Generated templates for 20 novel, expert-selected specialty topics and assessed clinical quality.

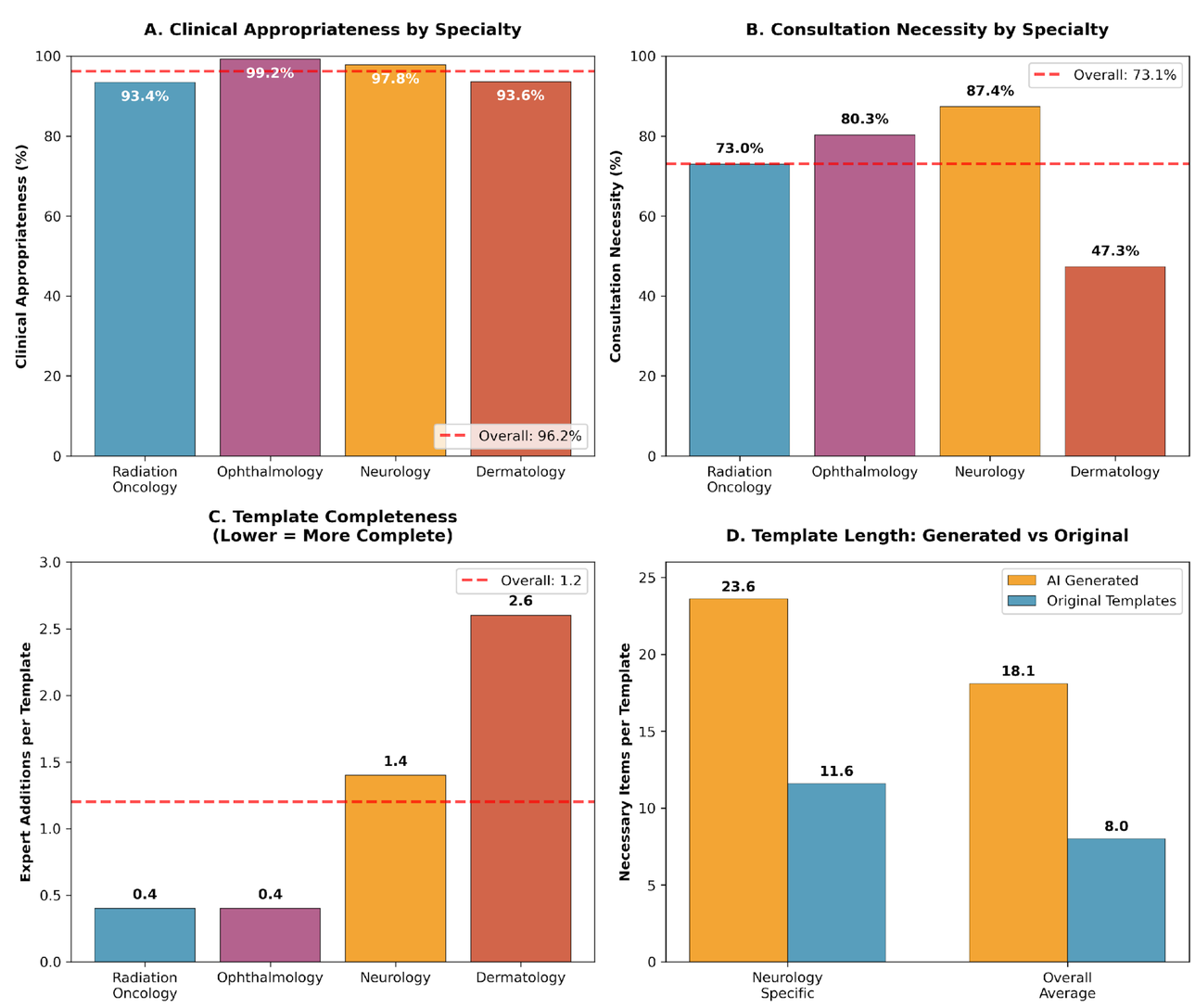

Figure 3. Expert Clinical Validation of Novel AI-Generated Medical Templates (A) Clinical appropriateness (% deemed appropriate by expert reviewer). (B) Consultation necessity by specialty (% deemed necessary by expert reviewer) (C) Template completeness (additional items added by experts per template). (D) Template length comparison between generated and original templates

Check Out the Paper!

Sign up to hear more about ARISE studies